

SHORT REPORT

Serum IgG levels in IV immunoglobulin treated chronic inflammatory demyelinating polyneuropathy

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► Additional material is published online only. To view please visit the journal online (<http://dx.doi.org/10.1136/jnnp-2012-304670>).

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Received 26 November 2012

Revised 14 February 2013

Accepted 3 March 2013

ABSTRACT

Objective To determine the variability of serum IgG in patients with chronic inflammatory demyelinating polyneuropathy (CIDP).

Methods All 25 CIDP patients had active but stable disease and were treated with individually optimised fixed dose IVIg regimens. IgG was measured by turbidimetry and variability was defined as coefficient of variation (CV).

Results The intra-patient variability of the pre-treatment IgG levels, post-treatment levels and increase in serum IgG shortly after IVIg (Δ IgG) was low (mean CV=3%, 4%, 10%). The inter-patient variability between patients treated with the same dose and interval was low in pre-treatment, post-treatment and Δ IgG level (mean CV=13%, 11%, 20%). The Δ IgG levels were associated with IVIg dosage ($r_s=0.78$, $p<0.001$).

Conclusions Clinically stable CIDP patients show a steady-state in serum IgG after serial IVIg infusions. The low intra- and inter-patient variability in IgG may indicate that constant levels are required to reach this stability.

INTRODUCTION

Intravenous immunoglobulin (IVIg) has been proven effective for Guillain-Barré syndrome (GBS) and chronic inflammatory demyelinating polyradiculoneuropathy (CIDP). The precise mechanisms of action are unknown, but the pleiotropic immunomodulating effects of IgG are assumed to be responsible for the therapeutic effect.¹ The optimum dosage and frequency of IVIg to reach a clinically stable situation in CIDP during maintenance treatment differs between patients and varies between 0.4–1.2 g/kg bodyweight every 2–6 weeks.² Currently, the optimum regimen has not been defined and cannot be predicted and needs to be established empirically in clinical practice.^{2–3} The variation in the required dosage and frequency of administration might be partially explained by individual differences in catabolism of IVIg. The aim of this study was to determine the intra-patient and inter-patient variability of serum IgG levels in clinically stable but IVIg-dependent CIDP patients receiving fixed dose maintenance treatment of IVIg.

METHODS

All patients fulfilled the American Academy of Neurology criteria for CIDP and participated in a randomised controlled trial comparing freeze-dried

IVIg (Gammagard S/D) with a liquid preparation (Kiovig).^{4–5} All were treated in neuromuscular centres and the dosage and frequency of IVIg was determined by neurologists experienced in treating CIDP. Muscle weakness was defined by the Medical Research Council (MRC) sum score (range 0–60) and Vigorimeter, disability by the overall disability sum score (ODSS) and sensory dysfunction by the INCAT sensory sum score (ISS).⁵ Medical ethical approval and informed consent was obtained.⁵

Patients had active CIDP and worsening of symptoms following IVIg reduction within the year before the start of the trial, confirming IVIg dependency.⁵ All were treated according to their own individually optimised IVIg dosage and frequency prior to trial entry and these regimens remained constant throughout the trial. To establish the optimal regimen of IVIg, the dosage was increased to achieve the maximal clinical response and the infusion frequency was shortened when patients were experiencing end-of-dose symptoms and signs.⁶ Regular attempts to decrease the dose were made as recommended.⁶

Serum IgG concentration (g/l) was determined by turbidimetry. At total IgG levels of 9.0 and 21.5 g/l, the between-run coefficients of variation were respectively 1.6% and 2.6% and the within-run coefficient of variation was <1%.⁷ Prior to this study, we had established that peak serum IgG levels were reached 1 min after infusion, and remained stable for at least 30 min after infusion. In this study, IgG levels were determined in serum samples obtained immediately before and 5 min after every infusion. The peak increase in serum IgG after IVIg (Δ IgG) was defined as the IgG level after treatment minus the level just before treatment. The coefficient of variation (CV) was calculated as the ratio of the SD to the mean multiplied by 100 (%). High variability in drugs is generally defined as a CV $\geq 30\%$.⁸

The Δ IgG of both preparations was compared using Wilcoxon matched-pairs signed-rank test. Correlation was tested with Spearman correlation coefficients (r_s). Analysis was performed using SPSS V.17.0. Two-sided p values <0.05 were regarded significant.

RESULTS

Twenty-seven patients were originally included in the trial. One patient was excluded from this study

To cite: Kuitwaard K, van Doorn PA, Vermeulen M, et al. *J Neurol Neurosurg Psychiatry* Published Online First: [please include Day Month Year] doi:10.1136/jnnp-2012-304670

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because of an unusual treatment regimen potentially influencing IgG levels (every other infusion a double dosage) and another because of premature termination of participation. All had been treated successfully with maintenance IVIg before starting the trial (mean 5 years, range 5 months to 13 years).

The Δ IgG after Gammagard infusion was smaller than after Kiovig (median 6.1 g/l (IQR 5–9) vs 6.8 g/l (6–9), $p < 0.001$), which may in part be attributed to the lower IgG content in Gammagard (95%) compared to Kiovig (~100%). Because of these differences in IgG content and the higher number of Kiovig infusions throughout the trial, we focused on the analysis of the IgG values after Kiovig infusions, although a similar low variability in IgG levels was observed after Gammagard. The lowest serum IgG level reached prior to infusion was 9.70 g/l (mean 15.0 g/l; median 15 g/l IQR 13–17) and the minimum Δ IgG level was 3.7 g/l (mean Δ IgG 7.8 g/l; median 7 g/l IQR 6–9). After serial infusions, intra-patient variability was low in pretreatment IgG levels (mean CV 3%, median 4 IQR 3–5), post-treatment levels (mean CV 4%, median 4 IQR 3–4) and Δ IgG levels (mean CV 10%, median 8 IQR 6–12) (figure 1). Although somewhat larger than the intra-patient variability, the inter-patient variability was small in pretreatment IgG levels (mean CV 13%, median 8 IQR 4–28) and post-treatment IgG levels (mean CV 11%, median 5 IQR 3–20) as well as Δ IgG levels (mean CV 20%, median 14 IQR 9–28) between those patients receiving the same dose and frequency of Kiovig (N=17, figure 1, see online supplementary table S1). When we calculated the increase in serum IgG 2 weeks after IVIg in the 13 patients with a frequency of one infusion every 2 weeks, the delta IgG was very low and close to zero (mean 0.09 g/l, median 0.07 g/l, range –0.61 till 0.7 g/l), whereas it was much larger in GBS (mean 7.8 g/l) due to the use of a larger dosage in GBS than used in the maintenance IVIg treatment in our CIDP cohort. The 2 week level was unsuitable for this cohort, and therefore, the peak IgG levels were determined shortly after infusion.

The post-treatment IgG levels and Δ IgG levels were related to the IVIg dosage administered per infusion ($r_s = 0.405$, $p < 0.05$; $r_s = 0.78$, $p < 0.001$), but not to the infusion frequency. The total dosage per infusion required to reach a stable clinical state and Δ IgG did not correlate with age, sex, bodyweight, lean body

mass, muscle strength, disability or sensory dysfunction (see online supplementary table S2).^{5 7}

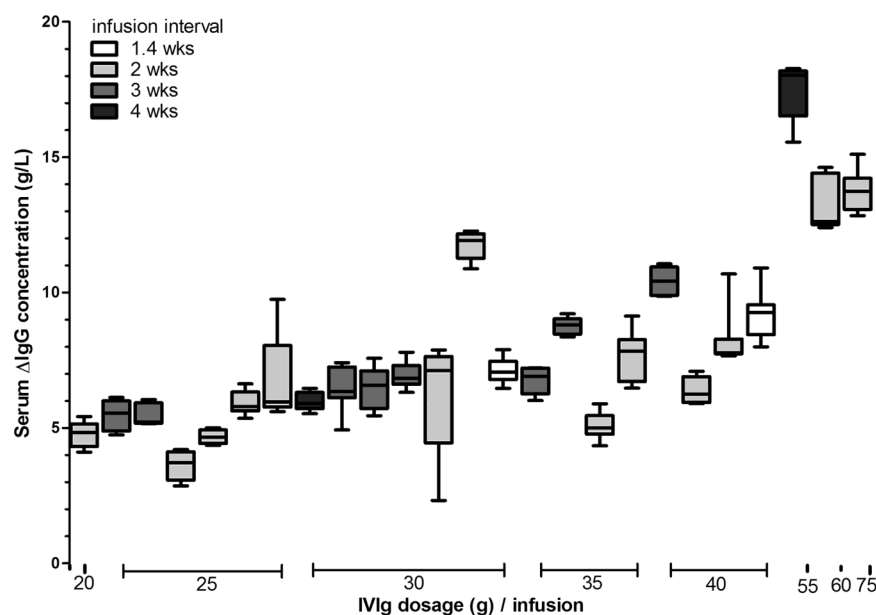
DISCUSSION

We showed that the serum IgG levels before and shortly after serial IVIg infusions were remarkably constant over time in patients with active but stable CIDP on constant maintenance treatment. This indicates that these patients have reached a steady state with a constant distribution rate and turnover of IgG without accumulation over time.

The dosage and frequency of IVIg required to maintain a clinically stable condition differs between CIDP patients, which might be due to interindividual differences in IVIg metabolism. Although there is some inter-patient variability between CIDP patients treated with the same IVIg dose and frequency, the mean CV can still be considered low from a pharmacological perspective, which leads us to a different conclusion than previously reported.⁹ A higher inter-patient variability in serum IgG levels 2 weeks after a standard course of IVIg has been observed in GBS (CV 31%) and primary immunodeficiency patients.^{7 10} This variation may depend on the activity of the disease, immunological host factors, baseline IgG levels, IgG glycosylation and Fc-receptor polymorphisms.^{3 11 12} The low inter-patient variability in serum IgG levels that we found in CIDP may be explained by the different study design in which none of the CIDP patients were treatment naïve, and all were already known to be IVIg responsive and clinically stable after a previously adjusted regimen of maintenance IVIg treatment. Variation in half-life of IgG is greater among patients with abnormal baseline IgG levels due to its concentration-dependent catabolism.^{13 14} The mean CV in baseline serum IgG level was somewhat lower in the CIDP patients treated with the same dose and interval (CV 13% N=17) than in GBS (mean CV 28% N=174) (Kuitwaard K, 2009, unpublished data), which might have contributed to the low variability seen in CIDP.⁷ Furthermore, the CIDP patients were treated with a lower IVIg dosage than the 2 g/kg used in GBS patients.

In patients with primary immunodeficiency, a minimum level of serum IgG may be required to prevent infections.¹⁰ In GBS, an increase of serum IgG level (about 7.30 g/l) 2 weeks after

Figure 1 Serum Δ IgG levels in patients receiving maintenance intravenous immunoglobulin (IVIg) treatment (Kiovig, N=25) Δ IgG=peak increase in serum IgG 5 min after IVIg infusion compared to pretreatment. Box and whisker plots show Δ IgG in 25 different patients, the box indicates 25th–75th percentiles; horizontal line indicates median value and the whiskers indicate minimum and maximum values. Patients are grouped by dosage. The colours of the boxes represent the infusion interval; patients receiving the same dosage and interval are displayed next to each other.



2 g/kg IVIg may be required for better recovery since the increase in the IgG level was independently associated with the ability to walk unaided at 6 months.⁷ The results of the current study suggest that a minimum serum IgG level and a minimum increase in serum IgG may be required to induce a clinical response and to reach a stable clinical condition in CIPD. This laboratory finding may be in line with the clinical observation that more than one IVIg course may be required to show improvement in CIPD.¹⁵ We did not include non-responsive or clinically unstable patients in this study; these patients may not have reached this minimum serum IgG level and may benefit from a higher IVIg dosage. Research in these patients is required to define if serum IgG levels can be used as a biomarker to monitor the effect of IVIg treatment.

No factors have been identified so far to predict the optimum regimen for maintenance IVIg treatment in CIPD.^{2–3} Bodyweight and the degree of disability were not related to the required dose of IVIg, confirming previous reports.² Factors other than bodyweight might determine the optimum dosage, and maintenance IVIg treatment can probably be started at a low dose and should only be increased if required by the clinical situation.²

The dose administered was the only factor related to the Δ IgG. The IVIg dosages or Δ IgG levels were not associated with bodyweight, lean body mass or severity of disease. In GBS, we demonstrated an association between disease severity and the increase in serum IgG level at 2 weeks after standard IVIg treatment.⁷ This difference may be explained by the fact that in the current study, all patients were in stable and good neurological condition being treated with optimised regimens.

We have shown that in active but stable CIPD, the inter-patient variability was larger than the intra-patient variability but still considered small. More studies are needed to determine whether unselected treatment-naïve CIPD patients do show a large variability in serum IgG levels after IVIg and if monitoring of serum IgG levels can be used to optimise IVIg treatment regimens in CIPD. Until such time, the reason why CIPD patients require different dosages in their IVIg maintenance treatment remains uncertain.

Acknowledgements The authors thank the patients and the nurses who took part in this study. Furthermore the authors thank the following contributors of the original trial: E Cats, MD (University Medical Center Utrecht, Utrecht, The Netherlands); SI van Nes, MD (Erasmus MC, University Medical Center, Rotterdam, The Netherlands); WCJ Hop (Erasmus MC, University Medical Center, Rotterdam, The Netherlands).

Contributors Study design: KK, PAvD, BCJ. Recruitment, data collection: KK, PAvD, MV, LHvdB, EB, AJvdK, W-LvdP, INvS, NCN, APT-G, WvR, TvG, BCJ. Statistical analysis: KK, PAvD, BCJ. Final report writing and approval: All.

Funding The original randomised controlled trial was supported by an unrestricted departmental research grant from Baxter Healthcare BV, The Netherlands. The pharmacokinetic studies were supported by the Netherlands Organization for Health Research and Development (907–00–111).

Competing interests KK was assigned to conduct a study comparing Gammagard S/D with Kiovig supported by an unrestricted departmental research grant from Baxter. KK received a speaker's fee from Grifols. PAvD received an unrestricted departmental research grant from Baxter to conduct a study comparing Gammagard S/D with Kiovig (the original trial) and to conduct a previous study comparing Gammagard S/D with or without methylprednisolone in GBS. PAvD received personal and departmental payments for consultancy/ RCT trial boards from Talecris, ZLB,

Baxter, Octapharma and received government research support from ZonMW, and the non-profit foundations Prinses Beatrix Spierfonds and Janivo Foundation, unrelated to this study. He serves as a member of the Cochrane Neuromuscular Disease Group editorial board. MV reports no disclosures. LHvdB received fees for lecturing and consultancy from Baxter BV. EB reports no disclosures. AJvdK reports no disclosures. W-LvdP received research support from the non-profit foundations Prinses Beatrix Fonds and Stichting Spieren voor Spieren, and travel grants from Baxter International. INvS received an unrestricted departmental research grant from Sanquin Blood Supply Foundation and from Actelion Pharmaceuticals Ltd; personal and departmental payments for lecturing and consultancy from Actelion Pharmaceuticals Ltd; he serves on scientific advisory boards for CSL-Behring and has received honoraria from CSL-Behring for consultancy. All consulting fees were donated to the stichting Klinische Neurologie, a local foundation that supports research in the field of neurological disorders. He received government research support the Netherlands Organization for Health Research and Development (940–33–024 and 903–51–201) unrelated to subject of publication; received support from the non-profit foundation Prinses Beatrix Fonds (MAR01–0213) and serves as a member of the Cochrane Neuromuscular Disease Group editorial board. N.C. Notermans reports no disclosures. APT-G reports no disclosures. WvR reports no disclosures. TvG reports no disclosures. BCJ received research government support from the Netherlands Organization for Health Research and Development (907–00–111) for the current project, and support from the Erasmus MC, Prinses Beatrix Spierfonds, Stichting Spieren voor Spieren, and GBS-CIPD Foundation International unrelated to subject of publication, and a travel grant from Baxter.

Provenance and peer review Not commissioned; externally peer reviewed.

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J Neurol Neurosurg Psychiatry published online March 28, 2013

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